Quick Revision Strategy for Data Engineering (3-5 Years) 2026

1. **Master the "Core 5" Concepts First (30 minutes)**

These are the first buckets interviewers use to judge maturity:

✓ Data Modeling

- Star vs Snowflake
- Fact vs Dimension
- Slowly Changing Dimensions (SCD Type 1–6)
- Normalization vs Denormalization
- When to use which (scenario-based)

Revision Hack:
Explain the above using *one project example*, e.g., *eCommerce orders pipeline.*

2. Pipelines & ETL/ELT Patterns (20 minutes)

Understand how to explain:

- Batch vs Streaming
- Incremental loads (high-value topic!)
- CDC (Change Data Capture)
- Idempotency
- Retry patterns
- Pipeline Monitoring & Alerts

Revision Hack:
Use a framework: **Source → Ingestion → Processing → Storage → Serving**.

3. Cloud Platform Deep Dive (AWS / Azure / GCP) (15 minutes)

Focus only on the services you use daily:

Azure

- ADF: Pipelines, Mapping Data Flow, Integration Runtime

- Databricks: Spark, Delta Lake
- ADLS Gen2
- Synapse / Fabric basics

AWS

- Glue, EMR, Lambda
- S3, Athena, Redshift
- Kinesis

Revision Hack:
Be ready to explain **one end-to-end architecture** you built.

4□. SQL + Performance Tuning (15 minutes)

80% of interviews test SQL.
Revise quickly:

- Window functions (ROW_NUMBER, LAG, LEAD)
- CTEs
- Joins vs EXISTS
- Query optimization (indexes, partitions)
- Explain query plans

Revision Hack:
Practice 3 classic questions:

1. Top N per group
2. Deduplicate records
3. Running totals

5□. Apache Spark Essentials (20 minutes)

Every 3–5 YOE role expects Spark experience.

Focus on:

- Transformations vs Actions
- Partitions & Shuffles
- Caching
- Wide vs Narrow transformations
- Delta Lake: MERGE, VACUUM, OPTIMIZE

Revision Hack:
Explain a **slow job** you optimized — interviews love this.

6□. Data Quality + Testing + Governance (10 minutes)

High demand topics for mid-level engineers:

- Data Quality Checks (Nulls, Constraints, Schema Drift)
- Great Expectations / Deequ
- Cataloging (Purview/Glue Data Catalog)
- Lineage

Revision Hack:
Be ready with **one example where bad data broke a pipeline**.

7. System Design for Data Engineers (15 minutes)

Be prepared for:

- How to design a daily batch pipeline
- How to design a near real-time streaming pipeline
- How do you scale pipelines?
- How do you make pipelines fault tolerant?

Revision Hack:
Use 4-step answers:
Ingestion → Storage → Processing → Consumption.

8. Behavioral + Scenario Answers (10 minutes)

They test your maturity with:

- How you debugged a failed pipeline
- How you handled schema changes
- How you optimized cost
- How you improved latency

Revision Hack:
Use **STAR format** (Situation → Task → Action → Result).

QUICK LAST-MINUTE REVISION

Know These Cold

- SCD Types
- CDC
- Window functions
- Spark partitions & shuffles
- Delta Lake MERGE
- ADF IR types / Databricks jobs
- Idempotent pipelines
- End-to-end architecture

- Data Quality checks
- SQL tuning
- Logging & Monitoring

If you can explain these **with real scenarios**, you will clear 90% of 3–5 YOE interviews.